

Course Project

[CMP175] - WEB PROGRAMMING

Major: Information Technology

Supervisor: Mr. Trịnh Công Nhựt

Student's name:

Full Name	ID:	Class:
Full Name	ID:	Class:
Full Name	ID:	Class:

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TOPIC 2. LANGUAGE CENTER WEBSITE

Part 1. Project Introduction (5 Points)

1.1 Problem Statement

Language centers commonly manage learners, teachers, courses, class schedules, registrations, placement tests, and tuition payments through separate spreadsheets, paper forms, messaging applications, and manual confirmation. This fragmented process makes information difficult to search, creates duplicate records, delays registration approval, and increases the risk of incorrect class capacity, payment status, or test results.

The problem is important because a language center serves several groups at the same time. Learners need accurate course information and a simple registration process; teachers need timely class lists and teaching materials; administrators need reliable operational data. When these groups work with disconnected information, the center spends more time correcting records and less time supporting learning quality.

The Language Center Management Website provides one shared web-based system. Guests can explore programs, classes, and teachers. Registered students can update profiles, register for classes, review payments, book placement tests, and submit consultation requests. Teachers can monitor assigned classes, view students, and manage class materials. Administrators can approve accounts and manage programs, classes, teachers, students, registrations, payments, and placement-test results.

- ☐ For learners: faster access to program details, class availability, registration status, payment information, and placement-test schedules.
- ☐ For teachers: a consolidated teaching dashboard, class roster, schedule overview, and document-upload area.
- ☐ For the center: centralized records, role-based workflows, operational statistics, reduced manual work, and more consistent service.

1.2 Topic Selection

The team selected this topic because language-center operations combine public information, user authentication, role-based authorization, database relationships, transactional workflows, file uploads, search, filtering, pagination, and dashboard statistics. It is therefore suitable for demonstrating the major learning outcomes of the Web Programming course while solving a recognizable real-world management problem.

Target users

- ☐ Guest: explores the center, programs, classes, and teacher information; creates an account and logs in.
- ☐ Student: maintains a personal profile, registers for classes, follows registration/payment status, books a placement test, and requests consultation.

- Teacher: views assigned classes and students, checks placement-test and consultation information, and manages learning materials.
- Admin: approves user requests and manages the center's complete operational dataset and dashboard.

1.3 Survey of Existing Websites

The survey was conducted on 9 June 2026 using the public home pages of VUS and Wall Street English Vietnam. The comparison focuses on information architecture and learner-facing functions rather than internal systems that are not publicly visible.

Website 1: VUS

Official website: <https://vus.edu.vn/>



Figure 1. VUS home page captured on 9 June 2026.

Aspect	Analysis
Advantages	Strong visual branding; clear course groups by age and learning need; prominent trial-test, support, and consultation actions; rich learning resources.
Disadvantages	The home page contains many promotional blocks and competing calls to action; key operational information such as class capacity, registration status, and payment progress is not presented as one simple workflow.
Features adopted	Program categories, visual program cards, teacher/center credibility, placement-test entry point, consultation call to action, and responsive navigation.

Website 2: Wall Street English Vietnam

Official website: <https://www.wallstreetenglish.edu.vn/>

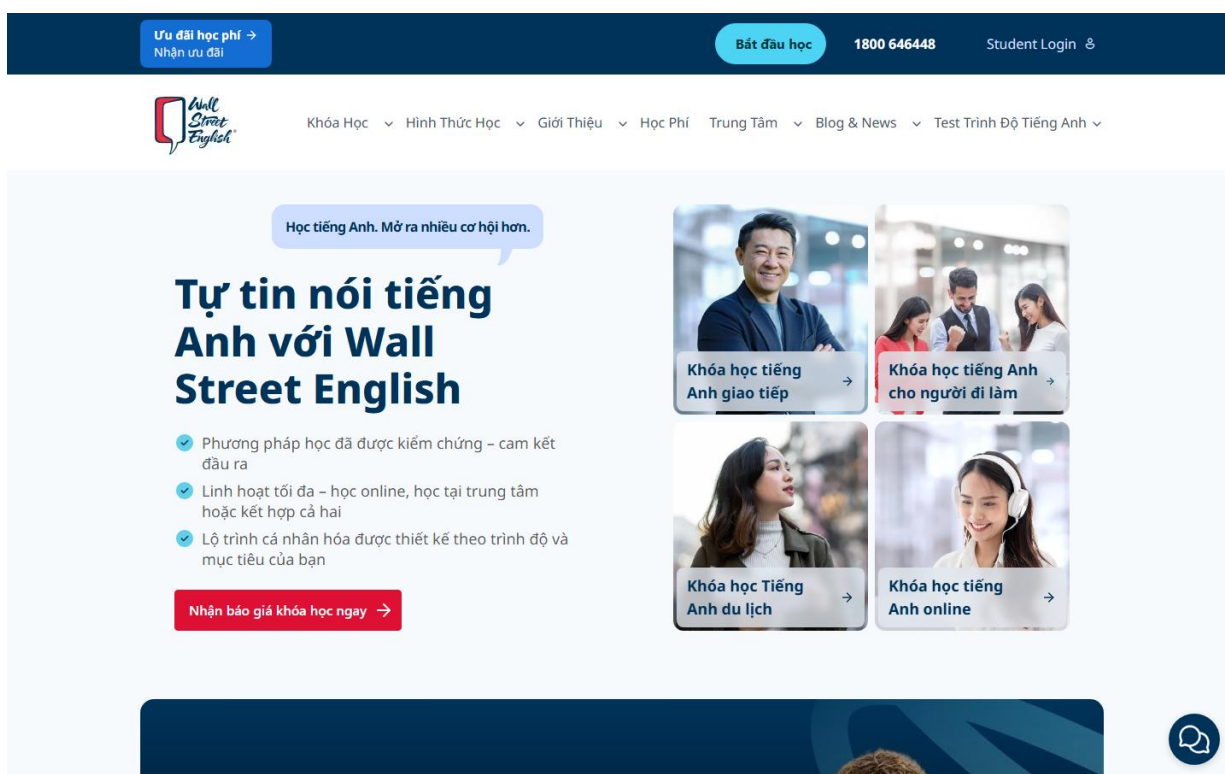


Figure 2. Wall Street English Vietnam home page captured on 9 June 2026.

Aspect	Analysis
Advantages	Clean hierarchy; clear value proposition; course cards for different learner goals; visible level-test and student-login links; strong consultation call to action.
Disadvantages	Course pricing and detailed class availability are not immediately visible; the public journey is consultation-oriented, so users cannot see a complete registration/payment workflow before contacting the center.
Features adopted	Simple hero message, goal-based program discovery, student login, level-test access, clear program detail pages, and focused calls to action.

Proposed feature set after the survey

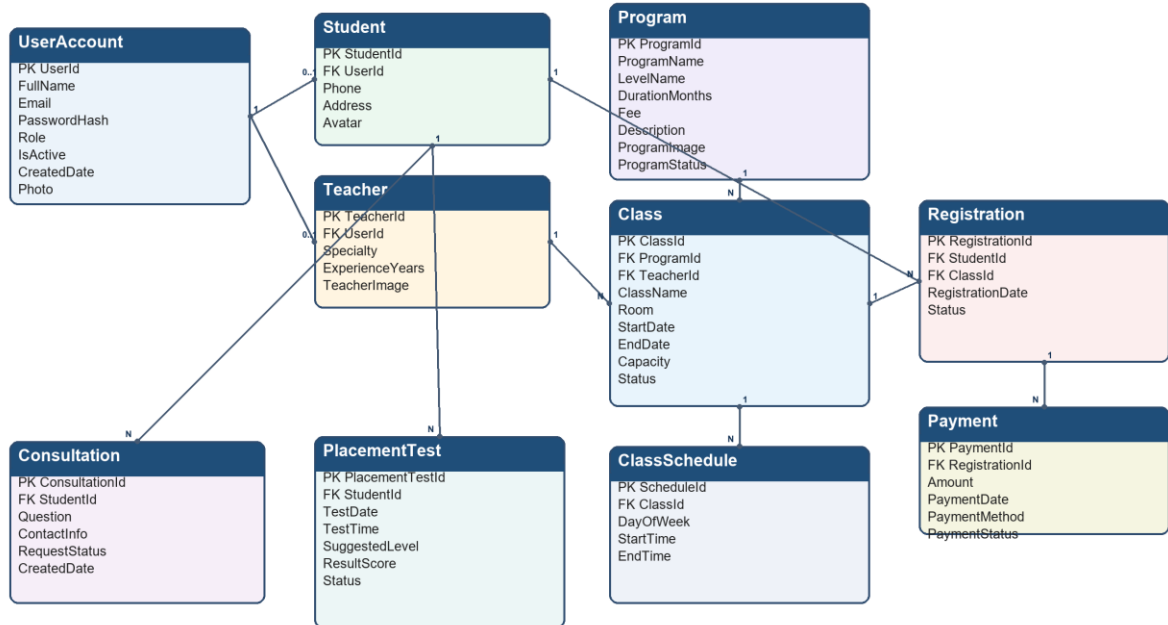
- ☐ Public home page with banner, statistics, featured programs, new classes, teachers, and clear Login/Register links.
- ☐ Program list with search, level filter, pagination, fee, duration, image, and detail page.
- ☐ Integrated account approval and role assignment instead of a purely promotional lead form.
- ☐ Student self-service for class registration, payment tracking, placement-test booking, and consultation requests.
- ☐ Teacher and Admin workspaces so the public website and internal operations share one database.

Part 2. System Design (15 Points)

2.1 ERD Diagram

LANGUAGE CENTER MANAGEMENT SYSTEM - ERD

PK = Primary Key FK = Foreign Key



Core process: User account → Student/Teacher profile → Program → Class → Registration → Payment

Supporting processes: Placement tests, consultations, and class schedules.

Figure 3. Entity Relationship Diagram of the Language Center Management System.

The database contains ten implemented entities. UserAccount stores authentication and role information. Student and Teacher extend a user account with role-specific profile data. Program and Class describe the academic offering. Registration links a student to a class, while Payment records tuition transactions for that registration. PlacementTest and Consultation support learner services, and ClassSchedule represents recurring teaching times.

Table	Primary key	Purpose
UserAccount	UserId	Authentication, identity, role, active status, and profile photo.
Student	StudentId	Student-specific phone, address, and avatar; references UserAccount.
Teacher	TeacherId	Teacher specialty, experience, and image; references UserAccount.
Program	ProgramId	Course name, level, duration, fee, description, image, and status.
Class	ClassId	Program offering assigned to a teacher, room, dates, capacity, and status.
ClassSchedule	ScheduleId	Day and start/end time for a class.
Registration	RegistrationId	Connects a student with a class and stores registration status.
Payment	PaymentId	Amount, date, method, and status for a registration.
PlacementTest	PlacementTestId	Test appointment, suggested level, score, and status.

Table	Primary key	Purpose
Consultation	ConsultationId	Student question, contact information, request status, and date.

Key relationships

- ☐ One UserAccount may have one Student profile or one Teacher profile, depending on the approved role.
- ☐ One Program has many Classes; each Class belongs to one Program.
- ☐ One Teacher may teach many Classes; each Class is assigned to zero or one Teacher during planning.
- ☐ Student and Class have a many-to-many relationship resolved by Registration.
- ☐ One Registration may have multiple Payment records to support transaction history or installments.
- ☐ One Student may create many PlacementTest and Consultation records.
- ☐ One Class may contain many ClassSchedule records.

2.2 Use Case Diagram

LANGUAGE CENTER MANAGEMENT SYSTEM - USE CASE DIAGRAM

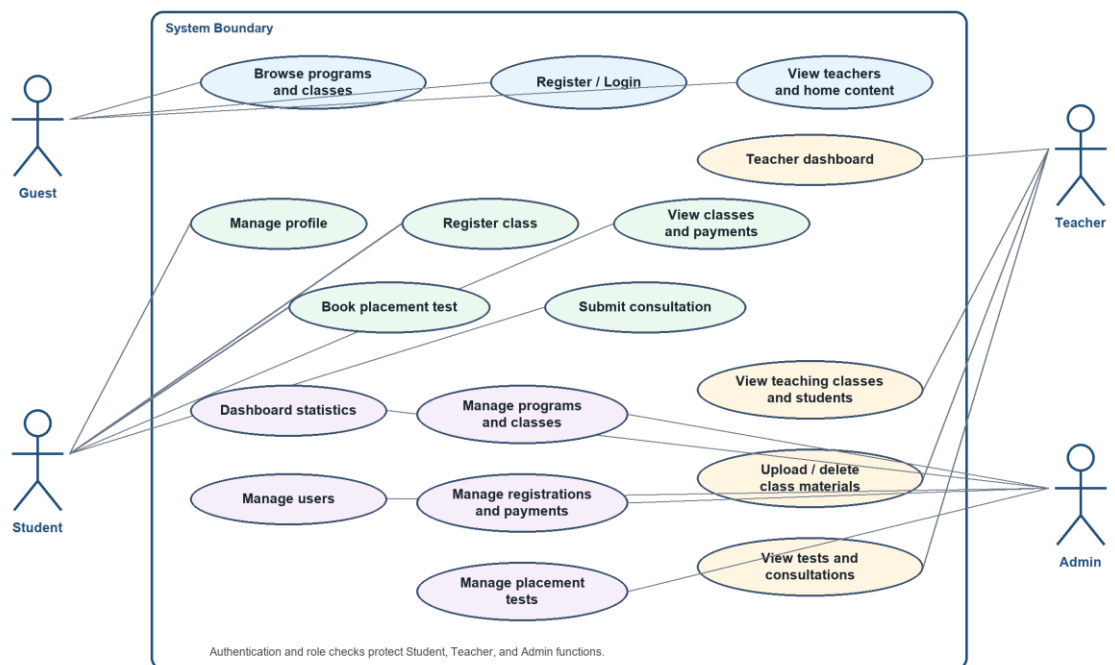


Figure 4. Use Case Diagram with Guest, Student, Teacher, and Admin actors.

Role	Main use cases
Guest	View home page, browse/search/filter programs, view program details and classes, view teachers, register, and log in.
Student	View/update profile, change password, upload avatar, register for a class, view classes and payments, book placement test, and submit consultation request.
Teacher	View dashboard, assigned classes and student lists, upload/delete class materials, view placement-test results and consultation feedback, and upload profile photo.
Admin	View dashboard statistics; approve accounts; manage students, teachers, programs, classes, registrations, payments, and placement tests.

Authorization is based on Session values (UserId and Role). Public actions are available to guests, while Student, Teacher, and Admin actions verify the current role before accessing role-specific data.

2.3 Sample Data

The following dataset is designed for demonstration and testing. It meets the minimum requirement: 5 programs, 10 classes, 10 students, 3 teachers, 20 registrations, and 10 placement-test records.

Programs (5 records)

ID	Program name	Level	Months	Fee (VND)	Status
P01	General English Foundation	A1	3	3,500,000	Active
P02	English Communication	A2-B1	4	4,800,000	Active
P03	IELTS Preparation	B1-B2	6	8,500,000	Active
P04	TOEIC Intensive	A2-B2	4	5,900,000	Active
P05	Business English	B1-C1	5	7,200,000	Active

Teachers (3 records)

ID	Full name	Specialty	Experience (years)
T01	Nguyen Minh Anh	IELTS / Academic English	8
T02	Tran Hoang Nam	Communication / TOEIC	6
T03	Le Thu Ha	Business / General English	10

Classes (10 records)

ID	Class	Program	Teacher	Room	Start	End	Cap.	Status
C01	GEF-A1-Morning	P01	T03	A101	01/07/2026	30/09/2026	20	Open
C02	GEF-A1-Evening	P01	T03	A102	06/07/2026	05/10/2026	20	Open
C03	COM-A2-Weekend	P02	T02	B201	04/07/2026	31/10/2026	18	Open
C04	COM-B1-Evening	P02	T02	B202	13/07/2026	13/11/2026	18	Open
C05	IELTS-5.5	P03	T01	C301	01/08/2026	31/01/2027	16	Open

ID	Class	Program	Teacher	Room	Start	End	Cap.	Status
C06	IELTS-6.5	P03	T01	C302	08/08/2026	07/02/2027	16	Planned
C07	TOEIC-550	P04	T02	B203	20/07/2026	20/11/2026	22	Open
C08	TOEIC-750	P04	T02	B204	27/07/2026	27/11/2026	20	Planned
C09	Business-B1	P05	T03	D401	03/08/2026	03/01/2027	18	Open
C10	Business-B2	P05	T03	D402	10/08/2026	10/01/2027	18	Planned

Students (10 records)

ID	Full name	Email	Phone	Address
S01	Pham Gia Bao	bao.pham@example.com	0901000001	District 1
S02	Nguyen Ngoc Lan	lan.nguyen@example.com	0901000002	District 3
S03	Tran Minh Khoa	khoa.tran@example.com	0901000003	Binh Thanh
S04	Le Hoang Yen	yen.le@example.com	0901000004	District 7
S05	Vo Thanh Tung	tung.vo@example.com	0901000005	Thu Duc
S06	Do My Linh	linh.do@example.com	0901000006	Go Vap
S07	Bui Quoc Huy	huy.bui@example.com	0901000007	Tan Binh
S08	Hoang Thu Trang	trang.hoang@example.com	0901000008	Phu Nhuan
S09	Dang Anh Tuan	tuan.dang@example.com	0901000009	District 10
S10	Mai Khanh Vy	vy.mai@example.com	0901000010	District 5

Registrations (20 records)

ID	Student	Class	Registration date	Status
R01	S01	C05	01/06/2026	Confirmed
R02	S01	C03	02/06/2026	Pending
R03	S02	C01	03/06/2026	Confirmed
R04	S02	C07	04/06/2026	Confirmed
R05	S03	C04	05/06/2026	Pending
R06	S03	C05	06/06/2026	Confirmed
R07	S04	C02	07/06/2026	Confirmed
R08	S04	C09	08/06/2026	Pending
R09	S05	C03	09/06/2026	Confirmed
R10	S05	C07	10/06/2026	Pending
R11	S06	C01	11/06/2026	Confirmed
R12	S06	C04	12/06/2026	Confirmed
R13	S07	C05	13/06/2026	Pending
R14	S07	C09	14/06/2026	Confirmed
R15	S08	C02	15/06/2026	Confirmed

ID	Student	Class	Registration date	Status
R16	S08	C03	16/06/2026	Confirmed
R17	S09	C07	17/06/2026	Confirmed
R18	S09	C09	18/06/2026	Pending
R19	S10	C04	19/06/2026	Confirmed
R20	S10	C05	20/06/2026	Pending

Placement tests (10 records)

ID	Student	Date	Time	Level	Score	Status
PT01	S01	15/06/2026	08:00	B1	62	Completed
PT02	S02	15/06/2026	09:00	A1	35	Completed
PT03	S03	16/06/2026	08:00	B2	78	Completed
PT04	S04	16/06/2026	09:00	A2	49	Completed
PT05	S05	17/06/2026	08:00	B1	65	Completed
PT06	S06	17/06/2026	09:00	A2	52	Completed
PT07	S07	18/06/2026	08:00	B1	-	Pending
PT08	S08	18/06/2026	09:00	A2	-	Pending
PT09	S09	19/06/2026	08:00	B2	-	Pending
PT10	S10	19/06/2026	09:00	B1	-	Pending

Data integrity rules used during testing: email addresses are unique; a student cannot register for the same class twice; class and program status values control availability; payment and registration statuses are updated only by authorized actions; foreign keys must reference existing records.

Part 3. Functional Requirements (70 Points)

Guest (10 Points)

- 1/ View Home (Banner, Programs, New Classes, Teachers, Navigation Menu, Login/Register Links)
- 2/ View Program List (Program Image, Program Name, Level, Duration, Fee, Search, Filter, Pagination)
- 3/ View Program Detail (Description, Output Standard, Fee, Duration, Related Classes, Register Button)
- 4/ Register (Full Name, Email, Password, Confirm Password, Validation)
- 5/ Login (Email, Password, Remember Me, Forgot Password)

Student (15 Points)

- 1/ My Profile (View Profile, Update Profile, Change Password, Upload Avatar)
- 2/ Register Class (Login Required, Class Selection, Registration Information, Confirmation)
- 3/ My Classes (Registered Classes, Schedule, Teacher, Registration Status)
- 4/ Payment (Payment Information, Amount, Payment Date, Payment Status)
- 5/ Placement Test Registration (Test Date, Test Time, Level, Status)
- 6/ Consultation Request (Question, Contact Information, Request Status)

Teacher (15 Points)

- 1/ Teacher Dashboard (Total Classes, Total Students, Teaching Schedule, Recent Activities)
- 2/ My Teaching Classes (Class List, Schedule, Room, Student Count, Class Status)
- 3/ View Class Students (Student List, Registration Date, Payment Status, Attendance Status)
- 4/ Manage Class Materials (Add Material, Edit Material, Delete Material, Upload Documents)
- 5/ View Placement Test Results (Student Name, Test Date, Suggested Level, Result)
- 6/ View Student Feedback (Feedback Content, Rating, Feedback Date)

Admin (15 Points)

- 1/ Dashboard (Total Programs, Total Classes, Total Students, Total Teachers, Total Revenue)
- 2/ Manage Programs (Add Program, Edit Program, Delete Program, Program Status)
- 3/ Manage Classes (Add Class, Edit Class, Delete Class, Assign Teacher, Class Status)
- 4/ Manage Teachers (Add Teacher, Edit Teacher, Activate/Deactivate Account)
- 5/ Manage Students (View Students, Edit Student Information, Activate/Deactivate Account)
- 6/ Manage Registrations (View Registration List, Update Status, Cancel Registration)
- 7/ Manage Payments (View Payments, Confirm Payment, Update Payment Status)
- 8/ Manage Placement Tests (Create Test Schedule, Update Result, View Test List)

Additional Features (10 Points)

- a) Email Notification (Account Verification, Class Registration Confirmation, Payment Confirmation)
- b) Online Payment (VNPay, MoMo, PayPal, Payment Confirmation, Transaction History)
- c) Deployment (Publish Website, Deploy to Hosting, Public Access URL)
- d) Dashboard Enhancement (Charts, Revenue Statistics, Student Statistics, Top Classes)

- e) File Upload (Profile Image, Teacher Image, Class Materials, Documents)
- f) Security Enhancement (Role-based Authorization, Password Encryption, Account Lockout)

UI/UX (5 Points)

- Responsive
- Validation
- Upload Image
- Error Handling

Part 4. Project Understanding Test (15 Points)

- a) Question 1. Database Understanding
- b) Question 2. Function Understanding
- c) Question 3. Basic Code Modification